

Tecnológico de Monterrey – CCM
Advanced Artificial Intelligence for Data Science I

Assignment: ETL and Prediction of Metro CDMX Ridership using Regression Models

Objective: Apply the ETL process (Extraction, Transformation, and Loading) to a real public-transit dataset and use supervised regression models to predict daily ridership. This file comes from an operational administrative pipeline and carries its defects. Expect to spend most of your effort on transformation, not on modeling.

Dataset

- **Name:** Afluencia diaria del Metro CDMX (desglosada por tipo de pago). **Source:** Portal de Datos Abiertos de la CDMX.
- **Size:** 1,174,095 rows × 7 columns, 2021-01-01 to 2026-06-30. **Columns:** fecha, mes, anio, linea, estacion, tipo_pago, afluencia, the last being the target.

Instructions

1. Extraction

- Load the dataset and report its shape, data types, missing values and duplicates.
- Report the number of unique values of every categorical column and **compare them against the real Metro network**. Document any discrepancy you find.

2. Transformation

- **Cleaning.** Resolve any inconsistency in how categories are represented. Define the natural primary key of the table and verify it holds.
- **Missing data.** The file has no NULLS, but a large share of rows are zero, and not for the same reason. Classify those zeros into at least three causes with evidence, then decide what to do with each and defend it. A zero produced by a station that was not operating is an absent measurement, not a low one.
- **Structural breaks.** The period covers real service disruptions and a change in fare products. Identify them from the data, and state whether you exclude, flag or model each one.
- **Feature construction.** Build calendar features, and join Mexican public holidays and daily weather (see annex). Audit them: measure whether each helps and report the result, **including features that turn out not to**.
- **Encoding and scaling.** Station is high-cardinality: compare at least two encoding strategies and justify your choice on statistical and memory grounds. Scale where the model requires it, fitting on the training split only.
- **Target transformation.** The target is strongly right-skewed. If you transform it, justify it and see experiment E3.

3. Loading

- Save the transformed dataset (CSV, Parquet or similar) and report the size before and after.
- Use that file as the sole input to modeling.

4. Modeling

The purpose here is not to find the best model, but to obtain an instrument good enough to test whether your ETL decisions were sound. Build it, then use it in step 5.

- **Define your unit of prediction and justify it.** You may model fare types separately or aggregate them. They differ by a near-constant ratio, so leaving them mixed lets a model score well by learning an accounting identity instead of demand.
- **Start with a mandatory baseline**, for example the historical mean per group. It is the reference against which everything else is judged; a model that does not beat it is not adding value, and reporting that is a valid conclusion.
- Implement **two** models: one from the linear family and one tree-based ensemble. Justify each choice by the assumption it makes and why that assumption is plausible here.

- **The train/test split must be chronological.** This is a time series. State your cut-off.
- Evaluate with MAE, MSE, RMSE and R^2 , and report training time and memory. Respect the computational constraints in the annex.

5. Controlled experiments

Run each experiment by changing **one** thing and holding everything else fixed. For each one, report the before and after metrics in a table, and explain the mechanism that produced the difference. The explanation is worth more than the number.

- **E1 – Random versus chronological split.** Refit your best model with a shuffled split. Report how much the metrics change, say which numbers you would show a client, and explain exactly what leaks.
- **E2 – With versus without your structural-break decisions.** Refit on the raw data, skipping the exclusions from step 2. Quantify what your ETL work was worth.
- **E3 – Transformed versus original scale.** If you transformed the target, report the metrics both ways. Explain why only one of the two is reportable, and what a reader would wrongly conclude from the other.

6. Evaluation and conclusions

- Select your best model and explain the criteria – accuracy alone is not sufficient. Discuss the limitations of MAE, MSE, RMSE and R^2 here, given the skew, the zeros, and how much variance station identity alone explains.
- Report feature importance for both models, comment on whether they agree, and on how correlated predictors may distort the ranking.
- Which stage of the ETL had the largest effect on your results, and what evidence supports it? Were the external joins worth their cost? Would you deploy this model to forecast next month, and what would break?

Deliverable

1. Format: Word or PDF for the report, plus the notebook as .ipynb. The notebook must run top to bottom without errors.

2. Minimum content: cover page (name, ID, course, assignment, date); explanation of the ETL applied with code snippets and results; documented list of the data-quality defects found and the decision taken for each; modeling process and results; **the three controlled experiments with their before/after tables and explanations;** conclusions.

Evaluation

Criterion	Points
Correct application of the ETL process	20
Data-quality findings and the decisions taken	15
Handling of structural breaks, with justification	10
Feature construction, and honest reporting of features that did not help	15
Modeling: baseline plus two models, chronological validation	12
Controlled experiments (Section 5), and the explanations given	15
Compliance with the computational constraints (annex)	5
Evaluation and conclusions	8
Total	100

Note: This assignment does not count directly toward the final evaluation, but a minimum of 80 points is required to unlock the Quiz. The Quiz grade is the one considered for the final module evaluation, and it will be conducted only in the classroom.

Submission deadline: Wednesday, August 19, 23:59 hrs. **Quiz date:** Thursday, August 20, during class.

See the **Technical Annex** for computational constraints, starter code and guidance on common failure modes.